

Tennis Shot vs. Movement Classification: A Comparative Analysis of Machine Learning Classifiers using Accelerometer Data

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Abstract—In sports analytics, distinguishing specific athletic actions from general movement is a critical challenge for automated performance tracking. This paper presents a machine learning pipeline designed to classify tennis shots (Forehand, Backhand, Serve) versus general court movement (Walking, Running, Idling) using smartwatch accelerometer data. We collected a dataset of approximately 1788 samples, segmented into non-overlapping windows, and extracted 18 statistical features per window. We evaluated five distinct classifiers: Decision Tree (DT), Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), and Multi-Layer Perceptron (MLP). Our experimental results demonstrate that the MLP model achieves the highest accuracy of 89%, effectively capturing the complex non-linear dynamics of tennis strokes compared to traditional linear models.

Index Terms—Activity Recognition, Machine Learning, Wearable Sensors, Accelerometer, Sports Analytics, Tennis

I. INTRODUCTION

The proliferation of wearable technology has opened new avenues for granular sports analytics, enabling athletes and coaches to monitor performance metrics in real-time. In tennis, tracking the frequency and quality of specific shots—such as forehands, backhands, and serves—provides valuable insights into a player’s strategy and workload. However, a significant challenge in developing automated tracking systems is the reliable differentiation between these specific stroke actions and the significant amount of background noise generated by general court movement, such as running to the ball or waiting for a serve.

The primary objective of this project is to develop a robust offline machine learning pipeline capable of distinguishing between “Tennis Shots” and “General Movement” using only tri-axial accelerometer data from a smartwatch. This binary classification task serves as a foundational step towards a more comprehensive multi-class shot recognition system. By filtering out non-shot activities, we can significantly reduce false positives in shot counting and analysis.

In this work, we implement a complete data processing pipeline, including data acquisition, window-based feature extraction, and model evaluation. We compare the performance of five standard machine learning classifiers to identify the optimal approach for this specific activity recognition task.

Our results highlight the superiority of neural network-based approaches in modeling the rapid and complex motion patterns associated with tennis strokes.

II. METHODOLOGY

Our system architecture follows a standard activity recognition pipeline consisting of data acquisition, preprocessing, feature extraction, and classification.

A. Data Acquisition

Data was collected using a smartwatch equipped with a tri-axial accelerometer, recording acceleration forces along the x , y , and z axes. The sampling rate was set to capture the high-frequency components of rapid tennis strokes. The dataset was labeled into two primary classes:

- **Class 1 (Shots):** Explicit tennis strokes including Forehand, Backhand, and Serve.
- **Class 0 (Movement):** Non-shot activities such as Walking, Running, and Standing (Idle).

The total dataset consists of approximately 1788 samples, split into a training set (80%, ~1430 samples) and a testing set (20%, ~358 samples), with a stratified split to maintain class balance.

B. Preprocessing and Feature Extraction

Raw accelerometer data is noisy and continuous. To prepare it for classification, we applied a windowing technique, segmenting the time-series data into non-overlapping windows of approximately 100ms. This window size was chosen to be sufficiently short to capture the instantaneous nature of a racket impact while providing enough data points to compute meaningful statistics.

For each window, we extracted a feature vector of size 18, consisting of six statistical metrics calculated for each of the three axes (x , y , z):

- 1) Mean
- 2) Standard Deviation
- 3) Maximum
- 4) Minimum
- 5) Root Mean Square (RMS)
- 6) Median

These time-domain features are computationally efficient and effective for characterizing physical movements. All models were trained on this full feature set.

C. Classifiers

We evaluated five distinct supervised learning algorithms to find the optimal decision boundary:

- 1) **Decision Tree (DT)**: A baseline model that offers interpretability but is prone to overfitting.
- 2) **Support Vector Machine (SVM)**: Effective in high-dimensional spaces, using a kernel to find a separating hyperplane.
- 3) **Logistic Regression (LR)**: A linear model that provides a probabilistic baseline.
- 4) **Random Forest (RF)**: An ensemble method that improves upon decision trees by reducing variance.
- 5) **Multi-Layer Perceptron (MLP)**: A feedforward artificial neural network capable of learning complex non-linear relationships.

D. Feature Selection

To reduce model complexity and identify the most predictive attributes, we applied Sequential Forward Selection (SFS) to our best-performing model (MLP). The objective was to maintain high accuracy while minimizing the feature count for potential real-time efficiency.

We successfully identified a reduced subset of just 3 features that retained approximately 86.87% accuracy (compared to 89% with the full set):

- 1) `acc_z_max`: Maximum vertical acceleration.
- 2) `acc_y_max`: Maximum forward/backward acceleration.
- 3) `acc_y_min`: Minimum forward/backward acceleration.

This analysis highlights that the Z-axis (vertical movement) maximum acceleration is the single most predictive feature for distinguishing tennis shots from general movement.

Implementation was performed using Python libraries *Pandas*, *Numpy*, and *Scikit-Learn*.

III. RESULTS AND DISCUSSION

A. Classifier Comparison

We evaluated the performance of each classifier on the held-out test set. The accuracy results are summarized in Table I.

TABLE I: Per-Class Performance Metrics of Classifiers on Test Set

Classifier	Acc.	Class 0 (Movement)			Class 1 (Shot)		
		Prec	Rec	F1	Prec	Rec	F1
MLP	0.89	0.88	0.90	0.89	0.89	0.86	0.88
SVM	0.88	0.85	0.94	0.89	0.92	0.82	0.87
Random Forest	0.87	0.88	0.87	0.88	0.86	0.86	0.86
Logistic Regression	0.85	0.86	0.84	0.85	0.83	0.85	0.84
Decision Tree	0.81	0.83	0.81	0.82	0.79	0.82	0.80

The Multi-Layer Perceptron (MLP) achieved the highest accuracy of 89%, followed closely by SVM (88%) and Random Forest (87%). The Decision Tree performed the worst with 81% accuracy, likely due to its inability to generalize well to the variations in human movement.

B. Analysis

The superior performance of the MLP suggests that the boundary between "shot" and "movement" in the feature space is non-linear and complex. Tennis shots involve dynamic, multi-axis rotations that simple linear models (like Logistic Regression) or axis-aligned splits (like Decision Trees) struggle to capture fully. The MLP's hidden layers allow it to learn abstract representations of these motion patterns.

Figure 1 displays the confusion matrices for all five classifiers. We observed that the most common error mode was misclassifying aggressive running as a shot, likely due to the high acceleration spikes associated with sudden stops and starts on the court. Conversely, soft touch shots (like volleys) were occasionally misclassified as movement due to their lower intensity.

IV. CONCLUSION

In this project, we successfully developed a machine learning pipeline to classify tennis shots versus court movement using wearable accelerometer data. Our comparative analysis demonstrated that a Multi-Layer Perceptron (MLP) outperforms traditional classifiers, achieving an accuracy of 89%. Furthermore, through Sequential Forward Selection, we demonstrated that a compact subset of just three features (dominated by Z-axis acceleration) can yield comparable performance (87%), suggesting a viable path for low-power embedded implementation.

Future work will focus on expanding the classification to identifying specific shot types (Forehand vs. Backhand vs. Serve) and implementing the model on an embedded device for real-time feedback.

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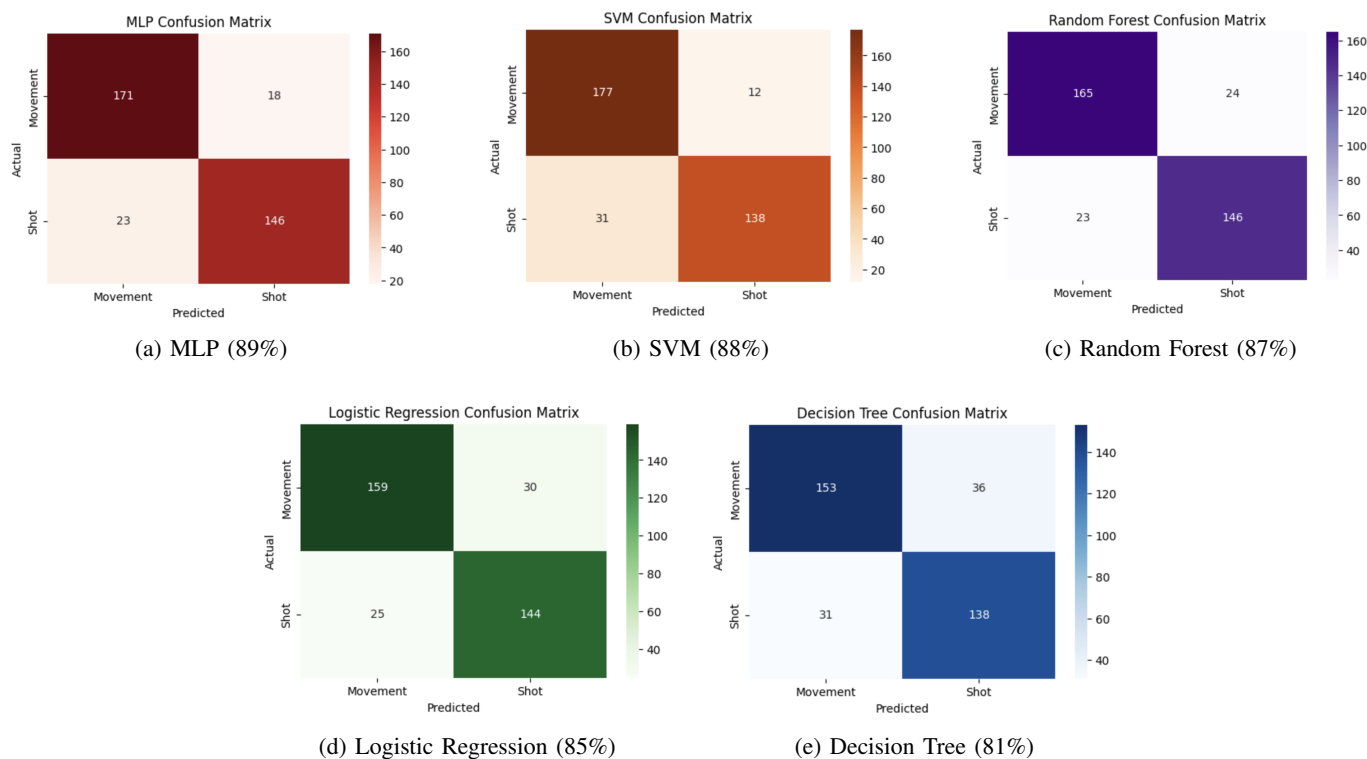


Fig. 1: Confusion Matrices for all five classifiers. The MLP demonstrates the highest diagonal density, indicating superior True Positive and True Negative rates.